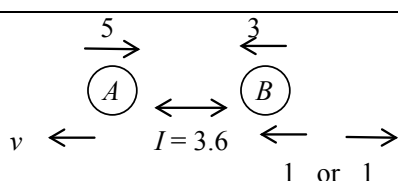
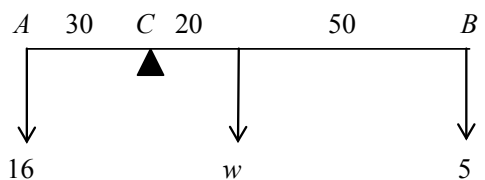
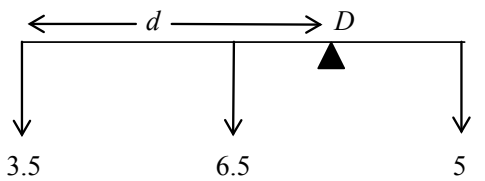


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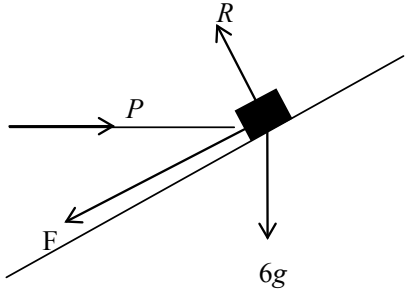
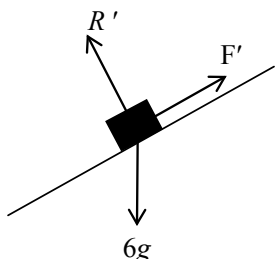
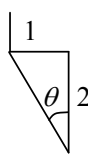
PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
1.	(a) $s = ut + \frac{1}{2}at^2$: $50 = 5 \times 4 + \frac{1}{2} \times a \times 4^2$ $\Rightarrow 30 = 8a \Rightarrow a = 3.75 \text{ m s}^{-1}$ (b) $30^2 = 5^2 + 2 \times 3.75 \times s$ $\Rightarrow s = 116\frac{2}{3} \text{ m}$	M1 A1 A1 (3) M1 A1 ft A1 (3) (6 marks)
2.	 Considering momentum of A: $3.6 = 0.5(5 + v)$ $\Rightarrow v = 2.2 \text{ m s}^{-1}$ Considering momentum of B: $3.6 = m(3 + 1)$ or $m(3 - 1)$ $m = 0.9$ or $m = 1.8$	M1 A1 A1 (3) M1 A1 (one) M1 A1 (both) (4) (7 marks)
3.	(a)  M(C): $16 \times 30 = w \times 20 + 5 \times 70$ (3 terms) $\Rightarrow w = 6.5 \text{ N}$ (b)  M(D): $3.5d + 6.5(d - 50) = 5(100 - d)$ $\Rightarrow d = 55 \text{ cm}$ (c) Tension equal along string, i.e. tensions = weights throughout or no contributions from strings in moments equation	M1 A1 A1 (3) M1 A2ft (-1 eeo) A1 (4) B1 (1) (8 marks)

(ft = follow through mark; -1eeoo = minus one mark for each error or omission)

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PROVISIONAL MARK SCHEME

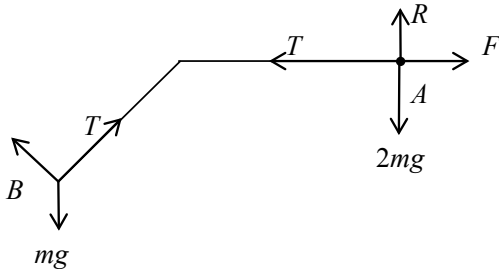
Question Number	Scheme	Marks
4. (a)	 $F = \frac{2}{5} R$ $R(\uparrow): R \cos 30^\circ - F \cos 60^\circ = 6g$ $R \frac{\sqrt{3}}{2} - \frac{2}{5} R - \frac{1}{2} = 6g$ $\Rightarrow R = 88.3 \text{ N (or 88 N)}$	B1 M1 A1 A1 (4)
(b)	$R(\leftarrow): P = R \cos 60^\circ + F \cos 30^\circ$ $= 74.7 \text{ N (or 75 N)}$	M1 A1 A1 (3)
(c)	 $\text{Component of weight } (\swarrow) = 6g \cos 60^\circ = 29.4 \text{ N}$ $R' = 6g \cos 30^\circ = 50.9 \text{ N}$ $F_{\max} = 0.4 R' = 20.36 \text{ N}$ $\text{Since } 29.4 > 20.36, \text{ the box moves}$	B1 M1 A1 M1 A1 cso (5) (12 marks)
5. (a)	 $\tan \theta = \frac{1}{2} \Rightarrow \theta = 26.6^\circ$ $\text{angle required} = 153.4^\circ$	M1 A1 A1 (3)
(b)	$\mathbf{a} = \frac{1}{3}[(\mathbf{i} - 2\mathbf{j}) - (-5\mathbf{i} + 7\mathbf{j})]$ $= (2\mathbf{i} - 3\mathbf{j}) \text{ m s}^{-2}$	M1 A1 (2)
(c)	$\mathbf{F} = m\mathbf{a} = 4\mathbf{i} - 6\mathbf{j}$ $ \mathbf{F} = \sqrt{16 + 36} = 7.21 \text{ N}$	M1 M1 A1 (3)
(d)	$\mathbf{v} = (-5 + 2t)\mathbf{i} + (7 - 3t)\mathbf{j}$	M1 A1ft (2)
(e)	$\mathbf{v} \text{ parallel to } \mathbf{i} + \mathbf{j} \Rightarrow \frac{-5 + 2t}{7 - 3t} = 1$ $\Rightarrow t = 2.4 \text{ s}$	M1 M1 A1 (3) (13 marks)

(cso = correct solution only)

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PROVISIONAL MARK SCHEME

Question Number	Scheme	Marks
6.		
(a)	shape (3, 2.5)	B1 B1 (2)
(b)	Area = $27 = \frac{1}{2} \times 1.5 \times 3 + 3T + \frac{1}{2} \times 2.5 \times 3$ $\Rightarrow T = 7 \text{ s}$	M1 A1 A1 (3)
(c)	shape $0 \leq t \leq 8.5$ shape $t > 8.5$ (2, 7 (ft), 2.5)	B1 B1 B1 (3)
(d)	(System) $T - 200g = 200 \times 2$ $\Rightarrow T = 2360 \text{ N}$	M1 A1 A1 (3)
(e)	(Man) $R - 80g = -80 \times 1.2$ $\Rightarrow R = 688 \text{ N}$	M1 A1 A1 (3) (14 marks)

Question Number	Scheme	Marks
7. (a)	 $R = 2mg \Rightarrow F = 2\mu mg$ $A: T - 2\mu mg = 2ma$ $B: mg \times \frac{1}{2} - T = ma$ Eliminating T: $3ma = \frac{1}{2}mg - 2\mu mg$ $a = \frac{1}{6}(1 - 4\mu)g (*)$	B1 M1 A1 M1 A1 M1 A1 (7)
(b)	$\mu = 0.2 \Rightarrow a = \frac{1}{30}g$ when string breaks: $v^2 = 2 \times \frac{1}{30}g \times h = \frac{1}{15}gh$ A decelerating with deceleration $f \Rightarrow 2mf = 2\mu mg$ $f = \mu g = \frac{1}{5}g$ Hence distance travelled during deceleration is given by $\frac{1}{15}gh = 2 \times \frac{1}{5}gd$ $\Rightarrow d = \frac{1}{6}h$ $\therefore$ Total distance = $\frac{7}{6}h$	B1 M1 A1 B1 M1 A1 cso (6)
(c)	Any two from: weight of pulley; friction at pulley; friction on slope; weight of string; string extensible; 'spin' of particle	B1 B1 (2) (15 marks)

((*) indicates final line is given on the paper; cso = correct solution only)